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Examining chatbot usage intention in a service encounter: Role of task complexity, communication style, and brand personality

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ABSTRACT

This study investigates the role of chatbot communication style (task vs. social oriented), task complexity (high vs. low), brand personality (sophisticated vs. sincere), and anthropomorphism on consumer trust and chatbot usage intention. Data is collected through three experiments conducted among US respondents (N = 328, 200, and 336). The results offer mixed insights as only one experiment supports that task complexity moderates the effect of communication style on trust, such that, task-oriented communication style of the chatbot leads to higher trust under high task complexity conditions. No significant differences in the moderating effect of task complexity on the relationship between communication style and trust is observed between sincere and sophisticated brands. Consistent across the three studies, it is observed that perceived anthropomorphism mediates the effect of communication style on trust which, in turn, affects intention to use the chatbot. The study contributes to literature on AI-enabled conversational agents, human computer interaction, anthropomorphism, and trust. Practically, the study offers insights for managers and service providers who wish to integrate chatbots and other AI enabled technology to enhance service delivery by providing efficient, cost-effective, and consistent support.

1. Introduction

Technological advancements have transformed the way companies carry out their business. One such advancement is the use of artificial intelligence (AI) at the service frontline that helps provide instant customer support through computer programs that simulate human conversations such as chatbots (Wirtz et al., 2018). These chatbots allow companies to efficiently cater to the needs of the consumers at any given time of the day without the presence of a real person such as a service agent or an employee (Cheng et al., 2021). With their ability to answer up to 80 % of frequently asked questions in a service encounter, chatbots can help improve customer satisfaction and engagement. As per global market insights, the chatbot market size was valued at $525.7 million in 2021 and is expected to grow at 25.7 % until 2030 (Grand View Research, 2022).

As service encounters between customers and service providers continue to evolve rapidly, there has been a recent surge in scholarly work that explores robot delivered frontline services (Forgas-Coll et al., 2023; Ding et al., 2024; Nguyen et al., 2023; McLeay et al., 2021). While some studies reveal that there is a general level of consumer enthusiasm or willingness to engage with AI agents, a significant amount of resistance has also been documented against AI enabled technologies especially in the service setting (Wang et al., 2023a, 2023b; Yang et al., 2023). This resistance has been linked to a lack of trust in chatbots (Chen et al., 2021). Consequently, without trust, chatbot adoption can be significantly hindered (Dekkal et al., 2023; Kaplan and Haenlein, 2019; van Pinxteren et al., 2019).

Prior studies have explored several factors that impact user's trust on and acceptance of a chatbot. These include chatbot-related factors: e.g., language style (Huang and Gursoy, 2024), communication style (Chen et al., 2021; Maar et al., 2023), chatbot appearance (Song and Shin, 2022), and visual cues (De Cicco et al., 2020); contextual factors: e.g., task complexity (Cheng et al., 2021), shopping task (Chen et al., 2021), and business type (Youn and Cho, 2023); consumer-related factors (Lee and Park, 2022; Maar et al., 2023), and brand-related factors (Liebrecht et al., 2021; Li and Shin, 2023). In the case of text based chatbots, specifically, the communication style is crucial in engaging with a user and gaining their trust. While extant studies have accounted for the role of communication style of the chatbot, little is known about how its effect on consumer trust is contingent on context specific factors, such as

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complexity of the task or information being sought from the chatbot. Recent work by Wang et al. (2023a, 2023b) contributed to this in the context of service recovery. They found that a social-oriented communication style leads to higher level of cognitive trust under high task complexity situations. However, a stream of work argues that task complexity may negatively influence the effect of chatbot friendliness on trust (Cheng et al., 2021). These contesting findings stress the need for continuing to discern how attributes related to the chatbot and the task interact to influence users' trust. Further, much remains to be known about brand related effects in trust building in chatbots.

Against this backdrop, the objective of this study is to understand how to promote consumers' trust and thereby intention to use the chatbot in customer service encounters. This study explores how the effect of communication style (task vs. social oriented) on trust is different under high (vs. low) task complexity conditions. In addition, the study examines a triple interaction effect of brand personality (sophisticated vs. sincere), task complexity and communication style on trust. Prior research has observed differences in consumers' response to service robot implementation among two different brand personalities (Choi et al., 2022). Therefore, it is predicted that the interaction effect of communication style and task complexity on trust is contingent upon this brand level distinction.

Another key construct driving consumers' trust formation and response in service encounters with chatbots is anthropomorphism which is defined as the process of ascribing human like mind and intentions to non-human entities (Rietz et al., 2019). It is argued that for some users, perceived human likeness of a chatbot can make it more controllable and relatable making the interaction easier and familiar for users. However, research has also found negative impacts of anthropomorphism on user trust and adoption intention where an anthropomorphized robot caused discomfort, strain, and threat to the user (Blut et al., 2021). Therefore, the role of anthropomorphism in influencing trust in chatbots warrants further investigation.

Overall, through three experiments the study investigates the following research questions: 1) How does chatbot communication style, task complexity, and brand personality interact to influence users' trust and thereby intention to use the chatbot, and 2) how does chatbot communication style influences user's trust through perceived anthropomorphism. Study 1 ($N = 400$) and study 2 ($N = 200$) test the proposed model using respectively a hedonic purchase context (i.e., a chatbot of a fictional hotel brand) and a utilitarian purchase context (i.e., a chatbot of a fictional laptop brand). Study 3 ($N = 400$) is extended to explore the role of brand personality using the context of two real-life hotel brands.

Together, the results advance our understanding of factors that influence consumer trust in AI-enabled chatbot and their usage intention. Theoretically, this research contributes to the discipline of AI-enabled conversational agents, human computer interaction, anthropomorphism, and trust. It advances existing knowledge about how chatbot related (Li and Wang, 2023; Roy and Naidoo, 2021), contextual (Hsu et al., 2023; Behera et al., 2024), and brand related factors (Yang and Hu, 2022) can interact to influence user's trust and thereby intention to use the chatbot. Practically, the study offers insights for managers and service providers who wish to integrate chatbots and other AI enabled technology in their service delivery model. Chatbots have the potential to enhance service delivery by providing efficient, cost-effective, and consistent support, which necessitates scholarly contribution to help drive innovation and inform best practices.

The next section provides an overview of the theoretical background for this research, followed by discussion of the hypotheses. Methods and results are discussed thereafter. Implications for theory, practice and future research are presented towards the end, followed by a brief conclusion section.

2. Theoretical background

2.1. Conversational agents and communication style

Research on AI-enabled technology and conversational agents has explored the role of design elements and agent attributes in driving their adoption. Attributes like voice characteristics (in case of voice-enabled agents), communication style, and non-verbal communication (in the context of text-based chatbots) are at the heart of their interaction with humans. Studies have explored the role of voice characteristics including accents and dialects (Torre et al., 2018; Andrist et al., 2015), culturally familiar conversation styles (Rau et al., 2009), and human vs. synthetic voices (Qiu and Benbasat, 2009) in building trust in voice enabled conversational agents. In the context of text-based chatbots, the quality and style of non-verbal communication replaces these attributes. Due to the absence of any physical presence in chatbots, its communication style can play an instrumental role in creating a successful encounter with the user (Go and Sundar, 2019).

Communication style can have multiple dimensions (van Pinxteren et al., 2023). Extant literature largely categorizes the communication style of a conversational agent as either social oriented or task oriented (Chattaraman et al., 2019; Dabholkar et al., 2009). A chatbot that uses a social oriented communication style is friendly, warm, and empathetic while a chatbot that is task-oriented concentrates on achieving a goal by using a tone this is precise and well structured around the main task at hand (Chen et al., 2021). A social oriented conversation is characterized by a casual or informal tone that exchanges social-emotional and affective information. When a digital assistant adopts a social oriented conversational style, it is programmed to exchange customary greetings, offer emotional support, indulge in small talk, and use more positive expressions to achieve social emotional goals (Kreijns et al., 2003). Task oriented communication or dialogue on the contrary is purely based on the task at hand and focuses mostly on achieving functional goals (Chattaraman et al., 2019; Elsholz et al., 2019). Service delivery literature has identified positive outcomes of using a social oriented communication style by service agents (Kim et al., 2011; Lloyd and Luk, 2011). Similarly, studies in human computer interaction have observed that social oriented communication style (De Cicco et al., 2020; Wang et al., 2023a, 2023b; Janson, 2023) can enhance user trust.

Extant studies have also established the role of chatbot communication style in determining service recovery satisfaction (Xu et al., 2022). Other forms of communication style that have been accounted for in chatbot literature include chatbot friendliness and empathy (Cheng et al., 2021), human like communication style (de Sá Siqueira et al., 2023), warm and competent (Schwede et al., 2022), and formal and informal language style (Li and Wang, 2023). In light of its pertinent role in influencing user experience, recent studies have called for continuing to expand the understanding of chatbot communication styles across various task and brand specific contexts (Janson, 2023; Li and Wang, 2023).

2.2. Anthropomorphism and conversational AI

Anthropomorphism is described as ascribing human like mind and intentions to non-humans (Epley et al., 2007) and is known to impact user engagement and acceptance of service robots. Consumer's propensity to anthropomorphize is particularly salient for social robots and smart technology (Novak and Hoffman, 2019). It is driven by their need to control and master a technology (Foehr and Germelmann, 2020; Epley, 2018; Waytz et al., 2010). The conversational nature of AI that characterizes social robots, chatbots, and voice enabled assistants further enhances the anthropomorphic cues that help replicate human-human interaction (Nass and Brave, 2005). Naming the chatbots also points to implicit anthropomorphism that facilitates consumers tendency to attribute human like mind and intention to technology (Foehr and Germelmann, 2020).

There is mixed evidence about the effect of anthropomorphism on human-technology interaction (Blut et al., 2021; Foehr and Germelmann, 2020; Rheu et al., 2021). By eliciting social presence anthropomorphism has been found to evoke favorable emotions about technology (van Doorn et al., 2017; Schuetzler et al., 2019). It also affects the consumers' trust in technology (Waytz et al., 2014), mitigates their risk perceptions (Kim and McGill, 2011); and favorably impact engagement with robots (Stroessner and Benitez, 2019) and chatbots (Tsai et al., 2021). There is also a stream of work that argues perceived human likeness of technology can have negative impact on consumers, as it may lead to unreasonable amount of emotional connection, trust, and enhance their expectations from the smart technology (Culley and Madhavan, 2013; Foehr and Germelmann, 2020). Other studies have found that consumers prefer to engage with a less human like robot (Broadbent et al., 2011) and high levels of anthropomorphism can negatively impact technology adoption (Luo et al., 2019). Hence, perception of anthropomorphism plays a complex and contingent role in determining consumer's usage intention of chatbots. Recent calls for research encourage the need for clear guidelines and its importance in driving crucial decisions for managers who wish to use AI to engage customers and enhance their experience (Blut et al., 2021; Huang and Rust, 2020).

2.3. Trust in AI enabled technology

Trust is defined as a cognitive process rooted in rational thinking and decision making (Kramer, 1999; Schoorman et al., 2007; Wilson et al., 2006). Scholars have defined cognitive trust as knowledge-based, shaped by evaluating the history, behavior, and prior experiences within the trustee-trustor relationship (Feng and Buxmann, 2020; McKnight et al., 2011; Wang et al., 2016). Mubarak and Petraite (2020) discuss "digital trust", defining it "the confidence of stakeholders on the competence of actors, technologies, and processes for establishing reliable and secure business networks" (p.3), which is crucial in driving collaboration between different actors. Recent literature has explored trust in AI enabled technology and introduced a new form of trust called "swift trust" that may apply to AI based conversational agents. Swift trust doesn't rely on historical interactions with the trustee (Jarvenpaa and Leidner, 1999). Instead, it is based on categorizing the trustee, forming a disposition towards that category, or finding similarities with other experiences, rather than depending on past interactions and experiences (Kramer, 1999).

Glikson and Woolley (2020) noted that AI transparency, tangibility and reliability can predict cognitive trust, while its anthropomorphism leads to emotional trust. Traditionally, trust reflects positive expectations that the object of trust will be useful, reliable, and competent. Such favorable expectations from the AI systems can be attributed to system specific features and assessment of its reliability, functionality, predictability and usefulness (Lockey et al., 2021). A systematic review on conversation agents by Rheu et al. (2021) identified social intelligence, voice characteristics and communication style, and anthropomorphic cues as key trust enhancing factors. Recent studies also identify social presence (Janson, 2023; De Cicco et al., 2020) and chatbot empathy and friendliness (Cheng et al., 2021) as key predictors of trust in chatbots. Literature focusing on factors impacting user trust in chatbot conclude trust in AI agents is a crucial indicator of acceptance (Gaudiello et al., 2016; Mostafa and Kasamani, 2022). Some scholars while comparing trust in AI vs human agents saw "algorithm aversion" where consumers were reluctant to use AI (Dietvorst et al., 2015) while others reported "algorithm appreciation" where users were seen putting more trust in AI compared to human agents (Logg et al., 2019). Further, scholars advocate expanding the understanding of trust in AI-enabled technology and conversational agents as they are being used in high stake situations and not just for entertainment or leisure purposes (Dignum et al., 2009; Rheu et al., 2021). Therefore, it is essential to discern the underlying mechanisms that lead to the development of trust in AI-enabled chatbots and its role in predicting users' behavioral intention.

Table 1 discusses the key contributions exploring chatbot adoption and engagement across varied contexts.

3. Hypothesis development

3.1. Effect of chatbot communication style and task complexity on trust

The communication style of a chatbot is known to greatly influence a user's experience and attitude towards the service (Elsholz et al., 2019). Researchers argue that the way a chatbot communicates or its interaction style or mannerism is a key component, and it must be consistent with the user's expectations and preferences both of which are known to differ with change in the needs of the user (Chattaraman et al., 2019). Prior studies offer evidence that social oriented communication style leads to favorable outcomes such as higher level of engagement (Huang and Ha, 2020; De Cicco et al., 2020). In a service setting warmth has been associated with showcasing service provider's friendliness and sociability, whereas competence reflects knowledge, skills, confidence, and efficiency (Kull et al., 2021; Kirmani et al., 2017; Tsai and Huang, 2002). Compared to other forms of conversational AI, e.g., voice enabled assistants, consumers engage with service chatbots for service assistance, product or service-related queries. Therefore, the nature of the relationship between customers and a chatbot is one between a consumer and a front-line employee.

Moreover, a user's response to a chatbot could change depending on the type of task at hand, or the service setting (Go and Sundar, 2019; Groom et al., 2009; van Pinxteren et al., 2023). Cheng et al. (2021) argue that in addition to chatbot attributes, task-based attributes can moderate the effect of chatbot attributes on user intention. They find that task complexity can negatively moderate the effect of chatbot empathy and friendliness on consumers' trust. Contesting this finding, a study by Wang et al. (2023a, 2023b) finds that consumers under high task complexity conditions exhibit higher cognition-based trust in chatbots that use a social oriented communication style. Overall, it is unclear how task complexity changes the effect of chatbot communication style on user's trust.

Complexity of the task enhances cognitive load and mental effort (Nguyen and Hsu, 2022) and may thereby influence user's perception of the chatbot's ability to handle the task. Xu et al. (2020) noted reduced perception of AI's problem-solving ability under high task complexity conditions. Similarly, Hsu et al. (2023) found that users preferred chatbots in the context of simple – one-attribute, information-light – contexts. For tasks involving complex information exchange, a casual conversation tone may reduce consumer's perception of competence and ability of the chatbot thereby reducing their trust. Conversely, for a complex task, a task-oriented communication style, which is target oriented and focuses on addressing the problem at hand, may be preferred by the user.

Therefore, we argue that the effect of communication style on trust is contingent on the complexity of the task. For a more complex task, consumers would prefer a more task-oriented tone, that signals competence and efficiency.

H1: When performing complex tasks, trust is greater for task-oriented (vs. social oriented) communication style.

3.2. Role of brand personality

Defined as a set of human characteristics ascribed to a brand (Aaker, 1997), brand personality has been a crucial predictor of positive attitudes and brand preferences in marketing and consumer behavior scholarship. The Big Five Brand Personality dimensions (Aaker et al., 2004) include Sincerity, Excitement, Competence, Sophistication, and Ruggedness. Brand personality reflects consumers' perception of the brand and can therefore influence key relations elements like trust. Prior research has established that brand personality dimensions, sincerity

Table 1
Overview of studies on chatbots.

Study	Study focus	Key findings
Huang and Gursoy (2024)	Examining the interaction between chatbot language style (concrete vs abstract) and customer decision making journey stage on user satisfaction with a chatbot and investigating the mediating role of social support	- Abstract language style in the early informational stage and concrete language in the transactional stage led to higher satisfaction levels. - Social support mediates language style and chatbot satisfaction by enabling users to form perceptions of information and emotional support.
Janson (2023)	Impact of anthropomorphic design features; personification (human like appearance) and social oriented communication style on social presence, satisfaction, trust and empathy	- Both design features have a significant impact on chatbot user perception, and they positively affect social presence. - Social oriented communication style can enhance customer satisfaction with a chatbot. - Social presence fully mediates both design features.
Li and Wang (2023)	Effect of communication style on continuance usage intention and brand attitude with the mediating role of parasocial interaction (PSI) and moderating role of brand affiliation.	- Informal communication style creates perception of parasocial interaction contributing to continuance usage intent and favorable brand attitude - Impact of chatbot language style reduces with no prior brand affiliation
Van Pinxteren et al. (2023)	Understanding engagement, rapport and service encounter satisfaction as mediators between communication style and adoption intention	- The presence of social cues significantly impact rapport with a chatbot in both hedonic and utilitarian contexts. - Chatbot mimicry of user communication style positively impacts engagement in a hedonic context. - Impact of engagement and rapport on adoption intent partially mediated by service encounter satisfaction
Jiang et al. (2022)	Examines the effect of social characteristics of chatbots on behavioral intentions	- Retailer experiential innovativeness, intimacy, and a sense of empowerment mediate the effect of chatbot social presence on consumer behavior. - Cute communication styles in low-level social presence effectively enhance relationships between social presence and experiential innovativeness. Increasing social presence levels with formal communication styles also strengthens these relationships.
Roy and Naidoo (2021)	Influence of anthropomorphic conversation style (Warm vs Competent) and time orientation (present vs future) on brand attitude and purchase intention.	- Matching chatbot's conversation style to user time orientation can help create brand preference. Social Brand perceptions mediate this relationship. - When chatbot uses warm conversation style brand with warm personality traits positively impact both present and future oriented individuals. Relationship is stronger when tested with present oriented people.
Soderlund et al. (2021)	Examining antecedents of perceived humanness (agency, emotionality and morality) and their impact on customer satisfaction with virtual agents	- The more users perceived the virtual agents to have agency, emotionality and morality the more users considered them to be human like. This relationship was positively linked to customer satisfaction with the virtual agent.
Cheng et al. (2021)	Examines the effect of chatbot empathy and friendliness on trust, and how task complexity and disclosure of the chatbot moderates this relationship	- Perception of chatbot friendliness and empathy positively influences trust. - Task complexity negatively moderates the relationship between friendliness and trust. - Chatbot disclosure negatively moderates the relationship between empathy and trust.
Youn and Jin (2021)	Examines the effect of consumer-chatbot relationship type (virtual assistantship vs. virtual friendship) on perceptions of brand personality, parasocial interaction, and customer relationship management	- Friendly customer-chatbot relationship increases para-social interaction. - Consumer-chatbot relationship type influences behavioral intention, satisfaction and trust, through competence brand personality - Parasocial interaction mediates the effect of customer-chatbot relationship for both competent and sincere brand personality.
De Cicco et al. (2020)	Effect of visual cues (avatar presence vs avatar absence) and interaction styles (social-oriented vs task oriented) on social presence, perceived enjoyment, trust, attitude	- Social oriented interaction style positively influences social presence, but avatar presence does not. - Social presence influences trust and perceived enjoyment, which leads to positive attitude towards the chatbot
Sheehan et al. (2020)	Relationship between Human-chatbot miscommunication and adoption intention	- Chatbot's ability to resolve miscommunication positively impacts adoption intent. - Anthropomorphism mediates this relationship - Higher the user's need for human interaction the stronger the mediation effect

and ruggedness are more closely associated with trust while excitement and sophistication influence brand affect more strongly (Sung et al., 2009; Sung and Kim, 2010). Therefore, the current study focuses on sophisticated and sincere brand personality to examine how the two dimensions may influence the effect of chatbot communication style and task complexity on trust. Sincere brands are perceived as down-to-earth, warm, friendly, family-oriented, honest, and reliable while sophisticated brands are associated with glamor, upper-class, elegance and charm (Aaker, 1997).

Recent research indicates that consumers show positive response to informal language style when used by warmer brands but expected formal communication style from competent brands (Leung et al., 2023). Gretry et al. (2017) also argued that consumers may expect brands with different personalities to use different communication styles. Competent brands may be expected to use a more formal style of communication whereas cheerful and exciting brands may be expected to use a more informal style of communication. In the context of social robot implementation, Choi et al. (2022) compared consumers response to adoption of high vs. low contact service robots by sincere vs. exciting brands. The study noted that consumers react negatively when sincere brands implement high contact service robots owing to poor perceived fit. Consumers expect qualities like warmth and friendliness from sincere brands. Conversely, use of casual conversation cues, like use of emojis or informal address by sophisticated brands may be perceived as inappropriate. As per Li and Shin (2023), use of emoticon by luxury brand chatbots reduces brand status perceptions that include sophistication and prestige.

Against this background, we argue that the interaction of task complexity and communication style will be moderated by brand personality. A task oriented and formal communication style under high task complexity will be deemed appropriate for a brand with sophisticated (vs. sincere) brand personality. A sophisticated brand will be expected to use a task-oriented communication style, more so when the task complexity is high because: a) a more formal and task-oriented communication style is more congruent with the brand personality, and b) high complex tasks demand a target-oriented and problem-focused communication style rather than a casual and informal communication style which might undermine the seriousness of the task at hand.

H2: For sophisticated (vs. sincere) brands, task oriented (vs. social oriented) communication will lead to higher trust in a high (vs. low) task complexity condition.

3.3. Effect of chatbot communication style and anthropomorphism on trust

Anthropomorphism is the process of making non- human subjects human like by ascribing them characteristics that resemble human beings. This is often done by designing them to give cues verbally or non-verbally that resemble a real person (Epley et al., 2007). As per the social response theory humans often interact with computers and conversational agents in the same manner in which they respond to humans (Nass and Moon, 2000). Anthropomorphizing chatbots influence how people interact with them as it often leads to a higher level of acceptance of service robots (Rietz et al., 2019). The innate human tendency to anthropomorphize objects was found to be more salient for objects like robots, and users anthropomorphize service robots more strongly than other types of technologies therefore it acts as a key construct while investigating user's response to AI based technologies such as chatbots (Duffy, 2003; Novak and Hoffman, 2019). Research suggests that anthropomorphic design of chatbots that may contribute to attraction and appeal could lead to emotional reactions which, in turn, impact trust in the technology (Chen and Park, 2021). Przegalinska et al. (2019) demonstrate that anthropomorphism and social cues can help create trust in human chatbot communication and that trust is a focal point of success in these interactions. The anthropomorphic and socially designed characteristics of AI chatbots can replicate a range of social cues encountered in human interactions. This replication might encourage users to attribute human qualities to their interactions with the AI chatbot, potentially initiating trust-building processes (Glikson and Woolley, 2020).

Research suggests that investigating the role of anthropomorphism leads to a better understanding of a user's tendency to apply social attributes to non- human agents (Verhagen et al., 2014). In line with the social response theory (Nass and Moon, 2000) an anthropomorphized agent will be able to foster a stronger relationship with the user. Prior studies have found that social oriented communication style enhances perceived social presence and enjoyment and has no direct effect on consumer's trust on the chatbot (De Cicco et al., 2020). While a chatbot takes on a social oriented communication style it will generate cues of humanlike communication such as showing more warmth, friendliness and expression which in turn can increase the feeling of social and personal interaction (Verhagen et al., 2014; Rietz et al., 2019). It may be argued that as social presence increases, users are more likely to associate social attributes such as trust and expertise with a conversational agent. Similarly, in the case of an interaction between a human and a chatbot where the chatbot uses a social oriented communication style the transmission of social cues would cause a perception of anthropomorphism in the user however this effect would be less prominent in a conversation where the chatbot uses a task-oriented communication style. Therefore, it is hypothesized:

H3: Perceived anthropomorphism mediates the effect of chatbot's social (vs. task) oriented communication style on trust, such that chatbot's social (vs. task) oriented communication evokes perceived anthropomorphism, which in turn increases user's trust.

3.4. Effect of trust on intention to use the chatbot

An important driver of usage intention is trust in an AI based technology. Trust is the willingness to use the technology while believing that it can fulfill its obligations, tell the truth and to act in the trustor's interest (Davenport, 2019). For environments that produce high levels of uncertainty (such as AI based systems the workings of which may not be clear to common users) trust plays a central role in forming a successful relationship with the user as the user decides to engage with the technology (Wirtz et al., 2018; van Pinxteren et al., 2019). Prior studies have documented a close relationship between trust and intention to use certain technology (Mostafa and Kasamani, 2022; Kim and Ko, 2012; Zhou and Tian, 2010; Liébana-Cabanillas et al., 2017). The effect of trust on usage intention has been documented in the context of voice-activated assistants (Pitardi and Marriott, 2021). Cheng et al. (2021) posited that trust in chatbot increases reliance and decreases resistance to the chatbot. In this vein, we argue that in a chatbot context, usage intention can be a reaction of trust in the technology. Therefore, the study proposes the following hypothesis:

H4: Trust in the chatbot will positively influence intention to use the chatbot.

Based on the above discussion, Fig. 1 depicts the proposed model.

4. Methods and results

4.1. Overview of studies

To test the proposed hypotheses, we conducted three experiments. Study 1 used the context of a hedonic purchase (hotel booking) and study 2 is based on a utilitarian purchase (buying a laptop). The objective of using a different context is to test whether the proposed relationships are specific to attributes related to the nature of purchase. To generate the stimuli, the authors searched for frequently asked questions about booking a hotel and purchasing a laptop. They also examined popular hotel and ecommerce websites to identify the information that is readily available versus that which is not. Finally, the authors interacted with chatbots in real time to accurately simulate the conversation with a chatbot and enhance the believability of the hypothetical scenarios used in the experiments.

Studies 1 and 2 used a 2 (task complexity: high vs low) x 2 (communication style: task vs social oriented) between-subject experiment design and tested H1, H3, and H4. Study 3 used a real service brand scenario and a 2 (task complexity: high vs low) x 2 (communication style: task vs social oriented) x 2 (brand personality: sophisticated vs. sincere) between-subject design to test H1, H2, H3, and H4.

Across the three studies, communication style of the chatbot (social and task oriented) was manipulated with formal and informal communication. The social oriented chatbot spoke frankly while using emojis (Chen et al., 2021) and the task oriented chatbot remained goal oriented and structured (Dabholkar et al., 2009) by not showing any emotion and remaining task specific during the interaction. Task complexity was manipulated by depicting two different types of tasks: low task complexity involved seeking information that the chatbot is likely programmed to respond automatically; high task complexity involved seeking information which either the chatbot is often not programmed to respond or requires time to retrieve the answer. All the scenarios are included in the Web Appendix.

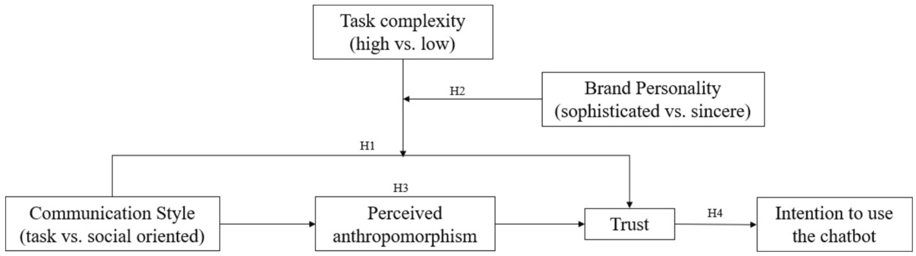

Fig. 1. Research Model.

4.2. Study 1

4.2.1. Methods

Before proceeding with the study, we conducted a pre-test to check the validity of the manipulation checks for task complexity and communication style. All participants were citizens of the United States of America and were recruited via Amazon Mechanical Turk (MTurk). For the pre-test, 56 US participants (Mean of age = 37 years; 56 % Male) were randomly assigned to one of the four scenarios. An independent sample *t*-test confirmed that there was a significant difference in perceived task complexity between the two conditions (high versus low task complexity) (MeanLC = 5.6, MeanHC = 6.3, $p < 0.05$). All respondents correctly identified the social versus task-oriented communication style, indicating that manipulation was successful.

Four hundred US participants were recruited via MTurk to participate in study 1. Participants were assigned to one of the four scenarios 2 (task complexity: high vs low) x 2(communication style: task vs social oriented) and asked to imagine that they are planning a vacation and are looking for information about a hotel. They visit the hotel's website and find an instant messenger service that can help answer questions about the hotel. Respondents were shown scripts of hypothetical conversations with a chatbot.

Manipulation check for communication style was conducted as per the study by Chen et al. (2021) by asking respondent to indicate true/false for the two statements: "The chatbot used an informal and casual conversation style with emojis"; and "the chatbot used a formal conversation style without emojis". This is in line with previous studies on AI-based technology adoption, (Chen et al., 2021). To check for manipulation of task complexity, we measured the variable using three items taken from Morgeson and Humphrey (2006) (e.g., *The information being communicated with the chatbot is complex; the communication with the chatbot involves exchanging complicated information; the information being sought from the chatbot is relatively difficult*).

Of the 400 respondents, 72 failed to pass the manipulation check on communication style, resulting in a final sample of 328. A one-way analysis of variance (ANOVA) confirmed a significant difference in task complexity between high and low task complexity conditions [$F = 31.420$, $p < 0.001$, Mean $_{\text{(high task complexity)}} = 2.59$ and Mean $_{\text{(low task complexity)}} = 1.93$], indicating that the manipulation of task complexity was successful.

The sample distribution for Study 1 includes 44.8 % female and 54.6 % male respondents. The age distribution reveals that the majority of respondents fall within the 30–39 years age group (37 %), followed by 40–49 years (24 %) and 18–29 years (16 %). See Table 2 for detailed information on the sample distribution according to age, gender, chatbot usage and purposes of using chatbots.

All measurement items for constructs were adopted from previous studies. Five items for anthropomorphism were borrowed from the work of Luo et al. (2019). Trust was measured using four statements borrowed from Cheng et al. (2021). Intention to use was measured using three items from Hill (1977). All scale items were measured using a seven-point Likert scale ranging from 1 (strongly disagree) to 7 (strongly agree). Communication style was coded as 1 for task-oriented communication style and 0 for social-oriented communication style. Task complexity was coded as 1 for high-task complexity and 0 for low-task complexity. Table 3 shows the descriptive statistics for the study's constructs. Scale item's reliability and validity for the constructs are summarized in Table 4.

4.2.2. Results of hypothesis testing

Trust. A 2 × 2 ANOVA indicated a significant two-way interaction between communication style and task complexity ($F = 4.078$, $p < 0.05$, Fig. 2). In keeping with H1, results from the ANOVA show that trust is higher for task oriented (vs. social oriented) communication style in high (vs. low) task complexity condition [Mean $_{\text{(task-oriented)}} = 5.86$; Mean $_{\text{(social-oriented)}} = 5.38$]. The main effects were also significant. Participants indicated higher trust for low (vs. high) task complexity [Mean $_{\text{(High task complexity)}} = 5.621$; Mean $_{\text{(low task complexity)}} = 6.092$, $p < 0.01$] and for task (vs. social) oriented communication style [Mean $_{\text{(task-oriented)}} = 5.984$; Mean $_{\text{(social-oriented)}} = 5.729$, $p < 0.05$].

To test H3, a mediation analysis was performed using Process v4.2 for SPSS 28 (Model 4) with 5000 bootstrap samples. Trust was the dependent variable, anthropomorphism was the mediator, and the focal independent variable was communication style. Respondent's age was entered as covariate to control for in the model. The indirect effect of communication style on trust through anthropomorphism was significant (indirect effect = − 0.3718, LLCI = −0.5358, ULCI = −0.2258), thus providing support for H3. In particular, the results indicate that a social-oriented communication style led to increased anthropomorphism ($\beta = -0.5549$, $p < 0.000$), and higher perceived anthropomorphism resulted in greater trust ($\beta = 0.6701$, $p < 0.001$).

The causal sequence from chatbot communication style (social vs. task oriented) to anthropomorphism, trust, and intention to use the chatbot (Fig. 1) was assessed using Process v4.2 (model 6) (Hayes, 2018). Intention to use the chatbot was the dependent variable, anthropomorphism and trust were the two sequential mediators. Respondent's age was entered as covariate to control for in the model. The overall path from communication style type to intention to use the chatbot through perceived anthropomorphism and trust was significant as the 95 % confidence interval (CI) did not include zero (indirect effect = −0.3681, LLCI = −0.5458, ULCI = −0.2082). Specifically, the results show that social-oriented communication style led to higher anthropomorphism ($\beta = -0.6380$, $p < 0.001$), and perceived anthropomorphism led to higher trust ($\beta = 0.5969$, $p < 0.001$). Trust positively influenced intention to use the chatbot ($\beta = 0.9668$, $p < 0.001$) providing support for H4.

4.3. Study 2

4.3.1. Methods

Participants for study 2 were recruited from Connect, a crowdsourcing platform by CloudResearch. All participants were citizens of the United States of America. Using the same process as study 1, participants were randomly assigned to one of the four scenarios 2 (task complexity: high vs low) x 2 (communication style: task vs social oriented). In this study participants were asked to imagine that they are

Table 2
Sample Characteristics.

Study 1 ($N = 328$)			
Gender		Age	
Female	44.8 %	18–29 years	16 %
Male	54.6 %	30–39 years	37 %
Non-binary/ third gender	0.6 %	40–49 years	24 %
Prefer not to say	0.0 %	50–59 years	13 %
		60–69 years	8 %
		70+ years	2 %
Chatbot use		**Purposes for chatbot use**	
Yes	94.21 %	Complaint resolution	54 %
No	3.66 %	Tracking an order	55 %
Can't remember	2.13 %	Online shopping	47 %
		Ordering food online	25 %
		Other	5 %
		Travel booking	13 %
		Financial advice	7 %
		Do not remember	2 %

Study 2 (N = 200)			
Gender		Age	
Female	36.50 %	18–29 years	22 %
Male	59.50 %	30–39 years	41 %
Non-binary/ third gender	3.50 %	40–49 years	21 %
Prefer not to say	0.50 %	50–59 years	11 %
		60–69 years	7 %
		70+ years	1 %
Chatbot use		**Purposes for chatbot use**	
Yes	97.5 %	Complaint resolution	60 %
No	1.0 %	Tracking an order	46 %
Can't remember	1.5 %	Online shopping	51 %
		Ordering food online	23 %
		Travel booking	11 %
		Financial advice	6 %
		Other	18 %
		Do not remember	2 %

Study 3 ($N = 336$)			
Gender		Age	
Female	49.4 %	18–29 years	22 %
Male	49.4 %	30–39 years	39 %
Non-binary/ third gender	1.2 %	40–49 years	21 %
Prefer not to say	0.0 %	50–59 years	11 %
		60–69 years	5 %
		70+ years	1 %
Chatbot use		**Purposes for chatbot use**	
Yes	96.1 %	Complaint resolution	63 %
No	2.7 %	Tracking an order	49 %
Can't remember	1.2 %	Online shopping	41 %
		Ordering food online	26 %
		Travel booking	9 %
		Financial advice	5 %
		Other	13 %
		Do not remember	3 %

planning to buy a new laptop (i.e., a utilitarian purchase) and are looking for information about certain features. They visit the hypothetical laptop brand's website and find an instant messenger service that can help answer questions about the product (Web Appendix). A pre-test with 80 US respondents (Mean of age = 30.3 years; 56 % Female) confirmed that the manipulation of task complexity (MeanLC = 2.25, MeanHC = 3.47, $p < 0.001$) and communication style was successful. Following the pre-test, 200 US participants were recruited for this study.

Table 3
Descriptive statistics for the study's constructs.

Study 1	N	Minimum	Maximum	Mean	Std. Deviation
Task Complexity	328	1	6	2.19	1.10
Trust	328	1.75	7	5.81	1.05
Anthropomorphism	328	1	7	5.38	1.18
Intention to Use	328	1	7	5.31	1.52
Study 2					
Task Complexity	200	1	7	3.04	1.51
Trust	200	2.5	7	5.61	1.07
Anthropomorphism	200	1	7	5.09	1.18
Intention to Use	200	1	7	4.87	1.60
Study 3					
Task Complexity	336	1	7	3.36	1.64
Sophisticated	336	1	7	5.24	1.26
Sincere	336	1.43	7	4.72	1.05
Trust	336	2	7	5.79	0.92
Anthropomorphism	336	1	7	5.43	1.074
Intention to Use	336	1	7	5.16	1.52

The measurement items and manipulation check questions were the same as used in study 1.

Manipulation check for communication style was successful, as all 200 participants correctly answered the questions ("The chatbot used an informal and casual conversation style with emojis" and "The chatbot used a formal conversation style without emojis"). A one-way analysis of variance (ANOVA) confirmed a significant difference in task complexity between high and low task complexity conditions [$F = 13.507$, $p < 0.001$, Mean $_{\text{(high task complexity)}} = 3.42$ and Mean $_{\text{(low task complexity)}} = 2.66$], indicating that the manipulation of task complexity was successful.

The sample distribution for Study 2 includes 36.5 % females and 59.5 % males. The age distribution reveals that the majority of respondents fall within the 30–39 years age group (41 %), followed by 18–29 years (22 %) and 40–49 years (21 %). Notably, 97.5 % of respondents reported using chatbots. The top three purposes for chatbot use among respondents are complaint resolution (60 %), online shopping (51 %), and order tracking (46 %) (see Table 2).

4.3.2. Results of hypothesis testing

4.3.2.1. Trust. Results of 2 × 2 ANOVA showed no significant difference in trust between task complexity*communication style ($F = 0.130$, $p = 0.718$, Fig. 3). Thus, H1 is not supported in study 2. Consistent with study 1, main effects show trust is significantly higher in task (vs. social) oriented communication style [Mean $_{\text{(task-oriented)}} = 5.769$; Mean $_{\text{(social-oriented)}} = 5.454$, $p < 0.05$]. The effect of task complexity is found marginally significant with trust higher for low task complexity condition [Mean $_{\text{(High task complexity)}} = 5.760$; Mean $_{\text{(low task complexity)}} = 5.464$, $p = 0.05$].

4.3.2.2. Anthropomorphism. The indirect effect from communication style to trust through anthropomorphism was significant (indirect effect = − 0.3292, LLCI = −0.5094, ULCI = −0.1443) thus providing support for H3. In particular, the results indicate that a social-oriented communication style led to increased anthropomorphism ($\beta = -0.5106$, $p < 0.003$), and higher perceived anthropomorphism resulted in greater trust ($\beta = 0.6448$, $p < 0.001$).

4.3.2.3. Intention to use the chatbot. The overall path from communication style to intention to use the chatbot through perceived anthropomorphism and trust was significant as the 95 % confidence interval (CI) did not include zero (indirect effect = −0.2602, LLCI = −0.4334, ULCI = −0.1154). Specifically, social oriented communication style led

Table 4
Scale items and reliability measures.

	Study 1		Study 2		Study 3	
Anthropomorphism		Factor loadings		Factor loadings		Factor loadings
The chatbot can converse like a human	**α = 0.870 CR = 0.896**	0.979	**α = 0.870 CR = 0.878**	0.75	**α = 0.887 CR = 0.889**	0.860
The chatbot appears to be happy	**AVE = 0.684**	0.928	**AVE = 0.591**	0.772	**AVE = 0.617**	0.811
The chatbot is friendly		0.837		0.876		0.702
The chatbot is respectful		0.623		0.682		0.695
The chatbot is caring		0.752		0.751		0.661
Trust						
The chatbot is honest and truthful	**α = 0.883 CR = 0.877**	0.777	**α = 0.884 CR = 0.886**	0.686	**α = 0.860 CR = 0.846**	0.736
The chatbot is capable of addressing the issues	**AVE = 0.704**	0.834	**AVE = 0.662**	0.782	**AVE = 0.648**	0.734
The chatbot's behavior and response meet my expectations		0.869		0.843		0.705
I trust the suggestions and decisions provided by the chatbot		0.902		0.924		0.866
Intention to use						
I would be interested in using this chatbot	**α = 0.946 CR = 0.951**	0.917	**α = 0.964 CR = 0.964**	0.953	**α = 0.968 CR = 0.966**	0.825
This chatbot increases my willingness to use it	**AVE = 0.865**	0.931	**AVE = 0.9**	0.952	**AVE = 0.906**	0.820
I will like to interact with this chatbot in the future		0.942		0.942		0.816

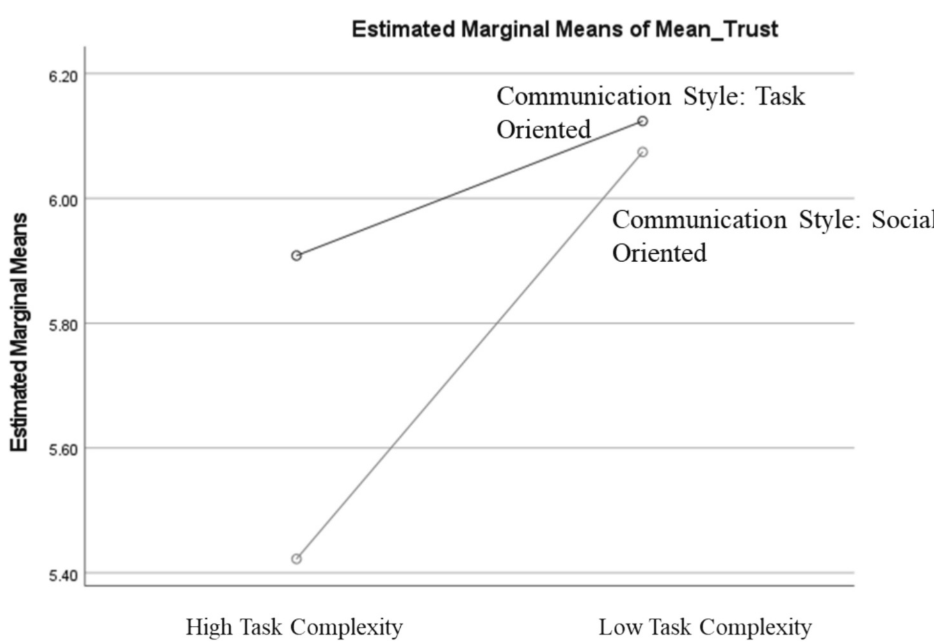

Fig. 2. Study 1 - Two-way interaction effect on trust.

to higher anthropomorphism ($\beta = -0.604$, $p < 0.001$), and perceived anthropomorphism led to higher trust ($\beta = 0.5890$, $p < 0.001$). Trust positively influenced intention to use the chatbot ($\beta = 0.7359$, $p < 0.001$) providing support for H4.

4.4. Study 3

4.4.1. Methods

To test the triple interaction effect of brand personality, communication style, and task complexity on trust, study 3 was designed using the context of a real hotel brand service chatbot. The Ritz Carlton Hotel Company and The Courtyard by Marriott were chosen to represent sophisticated and sincere brand personality respectively.

Before conducting the study, we performed a separate pretest to confirm the difference in brand personality of the two brands. Ninety six US respondents (mean age = 36.5, 35 % male, 65 % female) were randomly assigned to one of the two brands and asked to indicate the extent to which the brand appeared sophisticated (six items: upper class, glamorous, classy, elegant, charming, and timeless) (α = 0.939) and sincere (seven items: down-to-earth, family-oriented, honest, sincere, wholesome, friendly, reliable) (α = 0.895). The results suggested that the Ritz ($Mean_{Soph} = 5.78$) was perceived as more sophisticated than Marriott ($Mean_{Soph} = 4.53$) ($F = 35.235$, $p < 0.001$); and Marriott ($Mean_{Sinc} = 5.12$) was perceived as more sincere than the Ritz ($Mean_{Sinc} = 4.62$) ($F = 7.403$, $p < 0.01$).

Next, a total of 400 US participants were recruited from Connect, by CloudResearch and randomly assigned to one of the eight treatment conditions 2 (task complexity: high vs low) x 2 (communication style:

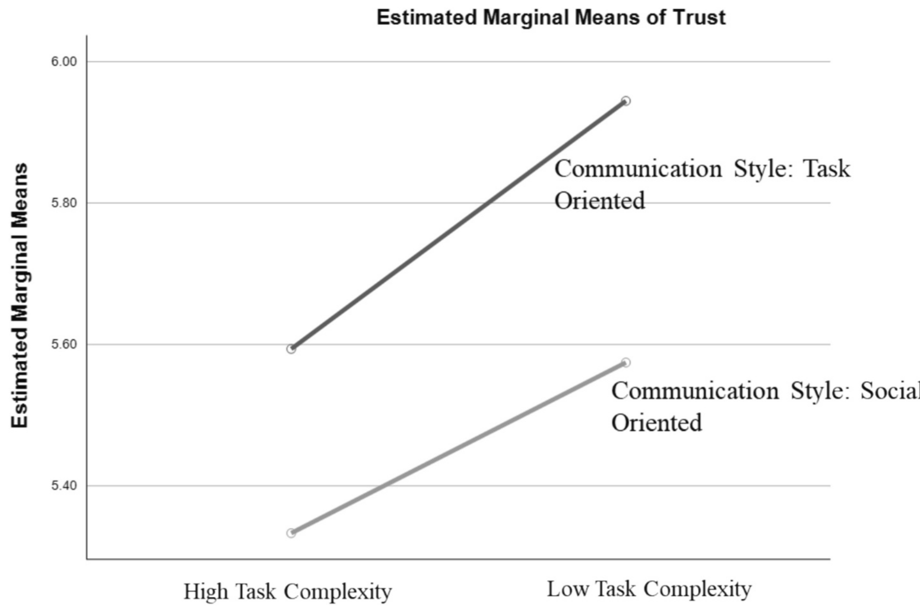

Fig. 3. Study 2 - Two-way interaction effect on trust.

task vs social oriented) x 2 (brand personality: sophisticated vs sincere). Participants were asked to imagine that they are planning to reserve a stay at The Ritz/The Courtyard by Marriott and are interacting with the brand's chatbot to seek some information related to their stay.

Sixty-four respondents failed the manipulation check for communication style as they did not correctly answer the questions: "The chatbot used an informal and casual conversation style with emojis" and "The chatbot used a formal conversation style without emojis". Their responses were removed from further analysis, resulting in a sample of 336 respondents. A one-way analysis of variance (ANOVA) found a significant difference in task complexity between high and low task complexity conditions [$F = 57.881$, $p < 0.001$, Mean $_{\text{(high task complexity)}} = 4.00$ and Mean $_{\text{(low task complexity)}} = 2.82$], indicating that the manipulation of task complexity was successful. Consistent with the pre-test, the Ritz ($\text{Mean}_{\text{Soph}} = 5.74$) was perceived as more sophisticated than Marriott ($\text{Mean}_{\text{Soph}} = 4.68$) ($F = 89.336$, $p < 0.001$); and Marriott ($\text{Mean}_{\text{Sinc}} = 5.12$) was rated as more sincere than the Ritz ($\text{Mean}_{\text{Sinc}} = 4.40$) ($F = 52.961$, $p < 0.001$).

The sample distribution for study 3 shows equal representation of female (49.4 %) and male (49.4 %) respondents. The age distribution shows a higher concentration of respondents in the 30–39 years age group (39 %), followed by 18–29 years (22 %) and 40–49 years (21 %). Regarding chatbot usage, 96.1 % of respondents reported using chatbot. Among the various purposes for chatbot use, the top three are complaint resolution (63 %), tracking an order (49 %), and online shopping (41 %) (see Table 2 for further details).

4.4.2. *Result of hypothesis testing*

4.4.2.1. Trust. To test H2 a moderated-moderation analysis was conducted using Process v4.2 for SPSS 28 (Model 3) with 5000 bootstrap samples. Trust was added as the dependent variable, communication style as the independent variables, task complexity as moderator 1 and brand personality as moderator 2. The three-way interaction was not significant (LLCI $= -0.9674$, ULCI $= 0.5789$, $p = 0.62$). Results of 2X2X2 ANOVA also showed no significant difference in trust between task complexity*communication style both for sophisticated and sincere brand ($F = 0.244$, $p = 0.621$, Fig. 4). Thus, H2 is not supported. H1 is also not supported as the interaction effect of communication style and task complexity was not significant ($F = 0.008$, $p = 0.93$).

Univariate analysis shows no significant difference in trust between task and social oriented communication style [Mean $_{\text{(task-oriented)}} = 5.812$; Mean $_{\text{(social-oriented)}} = 5.764$, $p = 0.629$]. The effect of task complexity is found significant with trust higher for low task complexity condition [Mean $_{\text{(High task complexity)}} = 5.569$; Mean $_{\text{(low task complexity)}} = 6.006$, $p < 0.01$].

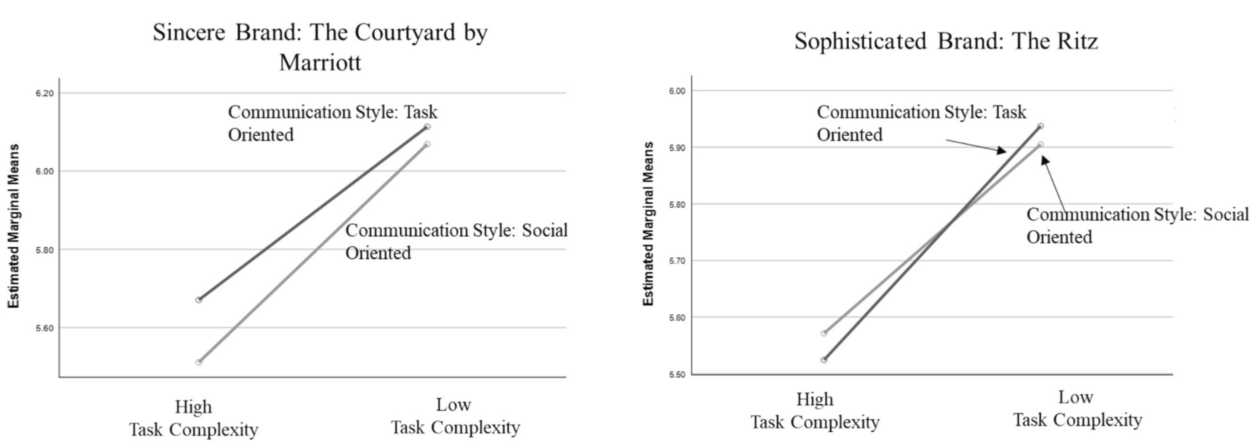

Fig. 4. Study 3 - interaction effect on trust for sophisticated vs. sincere brand personality.

4.4.2.2. Anthropomorphism. Consistent with study 1 and 2, the indirect effect from social oriented communication style to trust through anthropomorphism was significant (indirect effect = − 0.2952, LLCI = −0.4543, ULCI = −0.1586) thus providing support for H3. In particular, the results indicate that a social-oriented communication style led to increased anthropomorphism ($\beta = -0.4881$, $p < 0.000$), and higher perceived anthropomorphism resulted in greater trust ($\beta = 0.6049$, $p < 0.001$).

4.4.2.3. Intention to use the chatbot. The overall path from communication style type to intention to use the chatbot through perceived anthropomorphism and trust was significant (indirect effect = −0.1373, LLCI = −0.2178, ULCI = −0.0713). Specifically, social oriented communication style led to higher anthropomorphism ($\beta = -0.4986$, $p < 0.001$), and perceived anthropomorphism led to higher trust ($\beta = 0.5252$, $p < 0.001$). Trust positively influenced intention to use the chatbot ($\beta = 0.7993$, $p < 0.001$) providing support for H4.

5. Discussion

While extant studies have accounted for the role of communication style of the chatbot (Li and Wang, 2023; Roy and Naidoo, 2021; De Cicco et al., 2020), little is known about how its effect on consumer trust is contingent on context specific factors, such as complexity of the task and brand personality. This study extends the extant literature on text-based chatbots by exploring how the effect of communication style (task vs. social oriented) on trust are different under high (vs. low) task complexity conditions. In addition, the study examines a triple interaction effect of brand personality (sophisticated vs. sincere), task complexity and communication style on trust. Lastly, the study also examines the mediating role of anthropomorphism in the relationship between communication style and customer's trust. By addressing these research questions, the study provides the much-needed empirical evidence concerning the communication strategy of a chatbot and how it can play an instrumental role in influencing trust and usage intent.

Regarding the interaction effect of communication style and task complexity (H1), findings across the three studies show mixed results. In keeping with our prediction, results from study 1 reveal that in a high task complexity environment users put more trust in a text based chatbot that uses a task-oriented communication style compared to one that uses a social oriented. However, results from studies 2 and 3 indicate a lack of interaction effect of task complexity and chatbot's communication style.

In particular, for study 2, our results show that a task-oriented communication style always yields higher trust than a social-oriented communication style regardless of the level of complexity of the task. This result may be attributed to the different contexts used for experiments 1 and 2. As noted earlier, whereas experiment 1 is set around a hotel reservation (i.e., a hedonic purchase), the setting for experiment 2 is the purchase of a laptop (i.e., a utilitarian purchase). In this respect, it is argued that hedonic purchases are motivated by the desire for experiential consumption, fun, pleasure, and excitement, whereas utilitarian purchases are primarily practical, instrumental, and functional (Dhar and Wertenbroch, 2000). Accordingly, it seems plausible to assume that for utilitarian purchases, a text-based chatbot that uses a task-oriented communication style can be perceived as more competent in providing information than a chatbot using a social-oriented communication style, regardless of the complexity of the task.

Regarding study 3, the results point out an insignificant interaction effect of communication style and task complexity. Interestingly, in this study, the main effect of communication style on trust and the triple interaction effect of communication style, task complexity and brand personality (H2) are also insignificant suggesting that the type of language adopted by a chatbot is not a key determinant of consumer trust in the chatbot irrespective of the level of complexity of the task and/or the personality of the brand. We believe this result could be attributed to the fact that the two brands chosen for this experiment namely, The Courtyard by Marriot and the Ritz-Carlton, have very high levels of familiarity among the study's respondents (studies 1 and 2 did not allude to any particular brand). Thus, it is possible that the respondents' pre-existing relationships and experiences with those brands could have overshadowed or suppressed the effects of chatbot communication style, task complexity, and brand personality. This is in keeping with previous studies that reveal that strong relationships between brands and their customers can impact customers' beliefs and responses to the company marketing activities (Rog, 2014).

Finally, findings from our three studies confirm that a social-oriented communication style positively affects trust in a chatbot through perceived anthropomorphism. Findings demonstrate that when a text based chatbot uses a social oriented communication style the perception of anthropomorphism is stronger as compared to a task-oriented communication style. The mediation analysis confirms that perceived anthropomorphism mediates the relationship between communication style and trust. Higher trust, in turn, results in greater intention to use the chatbot. These findings give rise to important theoretical and practical implications, that are discussed next.

5.1. Theoretical implications

The current study theoretically contributes to the discipline of human computer interaction, trust building in AI-enabled technology, anthropomorphism, and technology adoption. By discerning the impact of communication style and its influence on user trust, the study expands the understanding of chatbot design attributes and their role in affecting relational outputs such as trust. While extant literature advocates the use of social oriented communication style, our results support that task-oriented communication style can lead to higher trust, pointing out the importance of task driven and competent communication by chatbots. This also validates an earlier finding by Hancock et al. (2011) who found that a robot's performance related factors have the biggest influence on trust followed by robot specific attributes like anthropomorphism and personality.

Academic studies have highlighted that trust is a fundamental factor influencing user acceptance and adoption of chatbot technology (Gaudiello et al., 2016). When users trust chatbots, they are more likely to engage with and use them for various tasks, which has practical implications for businesses and organizations implementing these AI-driven solutions. As text-based chatbots gain widespread application in high stake and complex tasks (Chong et al., 2021), understanding the drivers of consumers' trust in chatbots under high task complexity conditions becomes crucial for their long-term viability and success. The current study sheds key insights into this matter by examining the moderating effects of task complexity and brand personality on the relationship between chatbot communication strategies (task-oriented vs social-oriented) and customer trust. Regarding the moderating effect of task complexity, our results reveal a more complex relationship than previously assumed with task complexity moderating the relationship between communication style and trust only for hedonic purchases and not utilitarian purchases. Furthermore, findings from study 3 reveal that the moderating effect of task complexity is not contingent on brand personality perceptions (sincere vs. sophisticated). This is an original addition to the text-based chatbot literature where the role of brand related attributes has been rarely examined.

Furthermore, the study expands the growing body of knowledge on the implications of anthropomorphism in the context of text-based chatbots. With mixed findings about the role on anthropomorphism, there is a call for consistently exploring varied context and outcomes to establish the undermining effect of anthropomorphism and chatbot design cues (Blut et al., 2021; Adam et al., 2020; Go and Sundar, 2019). Prior literature has explored the impact of anthropomorphic design cues, including social oriented communication style, on social presence. The current study confirms that a conversational agent such as a text based chatbot when uses a social conversational language style causes an increase in the user's perception of anthropomorphism which positively impacts trust in the conversational agent.

Finally, the study adds to the discipline of technology adoption by integrating chatbot, task, and brand related attributes with trust and exploring its effect of intention to use the chatbot. Together, the three studies offer a nuanced understanding of the underlying mechanisms governing chatbot usage intention.

5.2. Practical implications

As consumers' use of chatbots to make hotel bookings and online purchases continues to increase, the findings of the study raise some pertinent implications for marketers and service providers. In particular, findings from this study suggest that customer's trust can lead to higher intention to use chatbots. Moreover, in order to increase trust in the chatbot, our findings reveal that task developers should focus on building a chatbot that communicates in a task-oriented communication style as this communication style translates into higher trust in the chatbot. Thus, the study's results reveal that in hedonic purchase contexts such as the booking a hotel room, for high complex tasks, a task-oriented communication style yields higher levels of trust than a social-oriented style, whereas under low task complexity, both task-oriented and social-oriented communication styles deliver similar levels of trust. Thus, it makes sense for companies to focus on building chatbots that use a task-oriented communication approach. Furthermore, our results show that in utilitarian purchase contexts, such as the purchase of a laptop, a task-oriented communication style also results in higher trust, irrespective of the level of complexity of the task.

The results also suggest that perceived anthropomorphism positively influences trust in chatbot and leads to higher intention to use the chatbot. Further it mediates the relationship between communication style and trust. A socially interactive chatbot can enhance consumer trust by evoking perception of anthropomorphism. By ascribing anthropomorphic cues to chatbots managers and designers can facilitate customer – chatbot interactions and enhance customer usage intentions as anthropomorphism can help create a sense of human connection with companies and brands.

5.3. Limitations and future directions

Overall, the results of this study should be interpreted in light of its intrinsic limitations. First, participants took part in hypothetical scenarios that involved a simulated conversation between a human and a chatbot, future studies could develop a real life chatbot and have the participants chat with the bot during the experiment. This would allow participants to fully experience the chatbot's conversation style and then provide real time evaluations based on that experience. This would not only provide more accurate results but would also give insights into other types of data such as how respondents respond to a social oriented chatbot versus a task oriented chatbot and if that has an impact on intention and actual usage behavior.

Second, the results of experiment 3 do not support the triple interaction effect of communication style, task complexity, and brand personality on trust. We attribute this result to possible contextual factors such as brand familiarity or prior customer-brand relationship. It is recommended for future research to continue exploring the role of these contextual factors to better understand the role of brand related factors in trust building with chatbots.

Third, the current study used two different settings, where respondents were encouraged to imagine that they used a chatbot for seeking information about a hotel's facilities and a laptop. Future research could reexamine these results in other use case scenarios such as banks, educational setting, social commerce, and online games or with brands that have different personalities.

Fourth, the study concerns a text based chatbot with no visual representation such as a profile photo, name, gender etc. but it would be interesting to see if adding these anthropomorphic cues would impact the results in anyway and how these additional chatbot characteristics evoke the perception of anthropomorphism. There are various aspects of a chatbot communication style that involves linguistics, adaptive personality, mimicking the user's conversation style that warrants further investigation for their impacts on user trust and usage intention. It would also be useful to verify if the same results apply to other speech-based assistants such as Google Home (smart speakers) or other digital assistants in kiosk settings or mobile phones and embodied conversational agents.

Lastly, this study did not take into account different aspects of anthropomorphism. Recent studies have reflected on the possibility of two types of anthropomorphism - mindless and mindful (Schuetzler et al., 2019; Araujo, 2018). Future studies can integrate these conscious and unconscious tendency to anthropomorphize non-human agents and investigate their role in determining chatbots effectiveness.

6. Conclusion

This study explores the effects of communication style (task vs. social-oriented) on trust under varying levels of task complexity and examines the interaction between brand personality (sophisticated vs. sincere), task complexity, and communication style on trust. Findings reveal that task complexity moderates the effect of communication style on trust in some contexts; specifically, a task-oriented communication style leads to higher trust under high task complexity in hedonic purchase settings. The results also show that perceived anthropomorphism mediates the relationship between social-oriented communication style and trust, which in turn affects the intention to use the chatbot. Interestingly, the triple interaction effect between communication style, task complexity and brand personality on chatbot trust was found to be insignificant. However, it is possible that respondents' preexisting familiarity with the brand chosen for the experiment may have overshadowed the expected effect. Overall, the study advances the understanding of factors influencing consumer trust in AI-enabled chatbots and provides practical insights for managers and service providers aiming to integrate chatbots into their service delivery models. The findings highlight the importance of task-oriented communication for building trust, especially in complex tasks, and the role of anthropomorphism in enhancing user trust and chatbot usage intentions.

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CRediT authorship contribution statement

Zara Murtaza: Conceptualization, Data curation, Formal analysis, Methodology, Writing – original draft. **Isha Sharma:** Conceptualization, Data curation, Formal analysis, Methodology, Supervision, Writing – original draft. **Pilar Carbonell:** Conceptualization, Formal analysis, Methodology, Supervision, Validation, Writing – review & editing, Writing – original draft.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.techfore.2024.123806.

Data availability

Data will be made available on request.

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